3.4.1 Newton's Method

Idea

- Assume that $f \in C^2[x_s, x_t]$ is a convex function (if f is not convex, the method can find only a stationary point, not a minimum point).
- Assume that an initial point $x^1 \in [x_s, x_t]$ is given and is sufficiently close to the minimum solution x^* with $f'(x^*) = 0$ and $f''(x^*) > 0$.
- Assume that $f(x^k)$, $f'(x^k)$ and $f''(x^k)$ are known at point x^k in iteration k $(k = 1, 2, \ldots)$.

- Construct a quadratic function which at x^k agrees with f up to second derivatives, i.e.,

$$f(x) \approx q(x) = f(x^k) + f'(x^k)(x - x^k) + \frac{1}{2} f''(x^k)(x - x^k)^2.$$

(It is easy to see that $q(x^k) = f(x^k), q'(x^k) = f'(x^k)$, and $q''(x^k) = f''(x^k)$).

- Let the next guess of the minimum point, x^{k+1}, be the minimum point of $q(x)$, i.e.,

$$q'(x) = 0 \Longrightarrow f'(x^k) + f''(x^k)(x - x^k) = 0 \Longrightarrow x^{k+1} = x^k - \frac{f'(x^k)}{f''(x^k)}.$$

- The process repeats until $|f'(x^{k+1})| < \varepsilon$ or other termination criterion is satisfied.

In fact Newton's method can be used to solve equation $g(x) = 0$ with the formula:

$$x^{k+1} = x^k - \frac{g(x^k)}{g'(x^k)}.$$

In a minimization problem, we replace $g(x)$ by $f'(x)$.

Advantage

- Fast if x^1 is near x^*. (need fewer iterations than the bisection method does)

Disadvantage

- Need second-order derivatives.
- The sequence $\{x^k\}$ may not converge to the solution x^* if the initial point x^1 is far away from x^*.

Algorithm

Step 1. Input $x^1 =$ the initial point. Set $k = 1$.

Step 2. Repeat the following formula:

$$x^{k+1} := x^k - \frac{f'(x^k)}{f''(x^k)}$$

until $|f'(x^{k+1})| < \varepsilon$ or other stopping criterion is met.

Step 3. Output the last x^{k+1} as x^*.

Geometric Meaning of Newton's Method.

See the figure on page 39. Suppose the current iterative point is x^{k-1}. We approximate the curve $f'(x)$ by its tangent line at the point $(x^{k-1}, f'(x^{k-1}))$. What we need to find is the intersection point x^* of the curve $f'(x)$ with x-axis, and we use the intersection point of the tangent line with x-axis to approximate x^*. The latter is just x^k according to the formula of Newton's method.

3.4.2 Secant Method

Idea

- Assume that $f \in C^2[x_s, x_t]$ is a convex function.
- Assume that two distinct initial points $x^0, x^1 \in [x_s, x_t]$ are given, and they are sufficiently close to the minimum solution x^* with $f'(x^*) = 0$ and $f''(x^*) > 0$.
- Assume that $f'(x^{k-1})$ and $f'(x^k)$ are known at iteration k $(k = 1, 2, \ldots)$.

- Construct a quadratic function:

$$\begin{aligned} q(x) &= f(x^k) + f'(x^k)(x - x^k) \\ &\quad + \frac{1}{2}\left[\frac{f'(x^k) - f'(x^{k-1})}{x^k - x^{k-1}}\right](x - x^k)^2. \end{aligned}$$

Note that $q(x^k) = f(x^k)$ and $q'(x^k) = f'(x^k)$, but $q''(x^k) \neq f''(x^k)$. In fact

$$q''(x^k) = \left[\frac{f'(x^k) - f'(x^{k-1})}{x^k - x^{k-1}}\right] \approx f''(x^k).$$

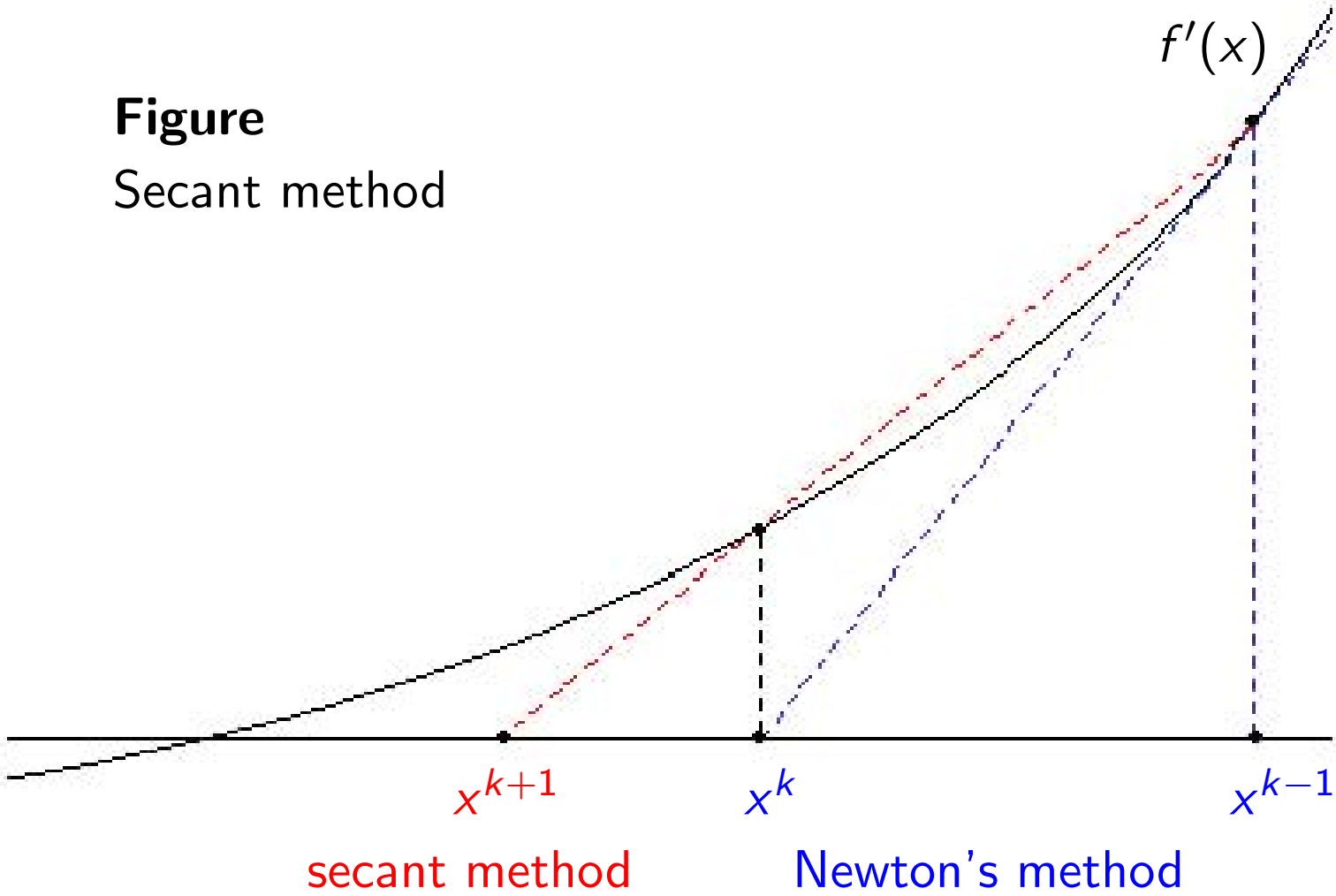
The geometric meaning is that in the graph of $g(x) = f'(x)$, the slope of the tangent line at x^k, i.e. $f''(x^k)$, is approximately replaced by the amount

$$\left[\frac{f'(x^k) - f'(x^{k-1})}{x^k - x^{k-1}}\right],$$

which is in fact the slope of the secant line passing through the two points $(x^{k-1}, f'(x^{k-1}))$ and $(x^k, f'(x^k))$ in the graph of function $f'(x)$. So, we used the secant line to replace the tangent line. That is the reason for the name of the method.

Figure

Secant method



- Let x^{k+1} be the minimum point of $q(x)$, i.e.,

$$x^{k+1} = x^k - \left[\frac{x^k - x^{k-1}}{f'(x^k) - f'(x^{k-1})}\right] f'(x^k).$$

Advantage

- Use first order derivatives only, and do NOT need to use second-order derivatives.

Disadvantage

- May be slower than Newton's Method.

Algorithm

Step 1. Input x^0, x^1 as two initial points.

Step 2. Repeat the following formula for $k = 1, 2, \ldots$

$$x^{k+1} := x^k - \left[\frac{x^{k-1} - x^k}{f'(x^{k-1}) - f'(x^k)}\right] f'(x^k)$$

until $|f'(x^{k+1})| < \varepsilon$ or other stopping criterion is met.

Step 3. Output the last x^{k+1} as x^*.

3.5 Speed of Convergence

- We often need to study how fast an algorithm converges to a minimum solution.
- **Definition.** Let the sequence $\{r_k\}$ converge to r^*. The *order of convergence* of $\{r_k\}$ is defined as the non-negative number p satisfying

$$0 < \lim_{k \to \infty} \frac{|r_{k+1} - r^*|}{|r_k - r^*|^p} = \beta < \infty.$$

- If the sequence has a convergence order p, we have asymptotically,

$$|r_{k+1} - r^*| \approx \beta |r_k - r^*|^p.$$

Note that if a sequence $\{r_k\}$ converges to r^* with a convergence order p', then p' must be unique. That is, $\{r_k\}$ cannot have two different convergence orders.

In fact if $\{r_k\}$ has converges order p', then

$$\lim_{k\to\infty} \frac{|r_{k+1} - r^*|}{|r_k - r^*|^{p'}} = \beta \quad (0 < \beta < \infty).$$

Then for any $p \neq p'$,

$$\begin{aligned} \lim_{k\to\infty} \frac{|r_{k+1} - r^*|}{|r_k - r^*|^{p}} &= \lim_{k\to\infty} \frac{|r_{k+1} - r^*|}{|r_k - r^*|^{p'}} \frac{|r_k - r^*|^{p'}}{|r_k - r^*|^{p}} \\ &= \begin{cases} 0 & \text{if } p < p' \\ \infty & \text{if } p > p' \end{cases} \end{aligned}$$

Therefore, p cannot be another convergence order.

Example 3.5.1:

- The sequence with $r_k = \frac{1}{k}$ converges to zero with order 1.
- The sequence with $r_k = a^k$, where $0 < a < 1$, converges to zero with order 1.
- The sequence with $r_k = a^{2^k}$, where $0 < a < 1$, converges to zero with order two.

Remark 3.5.1 Some convergent sequences may not have a convergence order. For example, in the above example, we know that

- sequence $r_k = \frac{1}{k}$ converges to 0 with order 1;
- sequence $r'_k = a^{2^k}$ $(0 < a < 1)$ converges to 0 with order 2.

Now if we mix the two sequences as

$$S = \{r_1, r'_1, r_2, r'_2, r_3, r'_3, \ldots\ldots\},$$

then the resulting sequence has no convergence order. In fact let the sequence be $S = \{s_1, s_2, s_3, \ldots\}$, then

$$s_{2k} = r'_k, \quad k = 1, 2, 3 \ldots$$

$$s_{2k-1} = r_k, \quad k = 1, 2, 3 \ldots$$

and

$$\lim_{k\to\infty} \frac{|s_{2k} - 0|}{|s_{2k-1} - 0|^p} = \lim_{k\to\infty} \frac{|r'_k|}{|r_k|^p} = \lim_{k\to\infty} \frac{a^{2^k}}{(\frac{1}{k})^p} = 0, \quad \forall p > 0,$$

$$\lim_{k\to\infty} \frac{|s_{2k+1} - 0|}{|s_{2k} - 0|^p} = \lim_{k\to\infty} \frac{|r_{k+1}|}{|r'_k|^p} = \lim_{k\to\infty} \frac{\frac{1}{k+1}}{(a^{2^k})^p} = \infty, \quad \forall p > 0.$$

Therefore, for any $p > 0$,

$$\lim_{k\to\infty} \frac{|s_{k+1} - 0|}{|s_k - 0|^p}$$

does not exist.

The following three convergence speeds are often met:

1. if $p = 1$ and $0 < \beta < 1$, then we say that r_k converges to r^* *linearly*;
2. if $p = 2$ and $0 < \beta$, then we say that r_k converges to r^* *quadratically*;
3. if $1 < p$, or if $p = 1$ and $\beta = 0$, then we say that r_k converges to r^* *superlinearly*.

Linear rate is a relatively slow rate. For a linearly convergent sequence $\{r_k\}$, when k is large, the errors are approximately

$$|r_{k+1} - r^*| \approx \beta |r_k - r^*|.$$

Especially, if β is close to 1, $\{r_k\}$ converges very slowly.

Quadratic convergence is very quick. We know that the third example above has order 2. If $a = \frac{1}{10}$, then the sequence $r_k = a^{2^k}$ becomes

$$\frac{1}{10^2}, \frac{1}{10^4}, \frac{1}{10^8}, \frac{1}{10^{16}}, \ldots$$

which tends to 0 very quickly.

For superlinear convergence, in the first case in the definition, as $p > 1$,

$$\frac{|r_{k+1} - r^*|}{|r_k - r^*|} = \frac{|r_{k+1} - r^*|}{|r_k - r^*|^p} \cdot |r_k - r^*|^{p-1} \to \beta \cdot 0 = 0,$$

which becomes the second case. Hence we can directly define superlinear convergence as

$$\lim_{k \to \infty} \frac{|r_{k+1} - r^*|}{|r_k - r^*|} = 0.$$

Superlinear convergence is considered as a quick convergence, even though it may not converge as fast as quadratic convergence.

Example 3.5.2 The sequence $\{r_k\}$ defined by $r_1 = 4$ and

$$r_{k+1} = 1 + (r_k - 1)/2^k, \quad \text{for } k = 1, 2, \ldots$$

i.e.,

$$\{r_k\} = 4, 2.5, 1.375, 1.0469, \ldots$$

meets the condition:

$$\lim_{k \to \infty} \frac{|r_{k+1} - 1|}{|r_k - 1|} = \lim_{k \to \infty} \frac{1}{2^k} = 0.$$

Hence $\{r_k\}$ converges to 1 superlinearly.

Remark 3.5.2 The sequence $\{r_k\}$ in Example 3.5.2 converges to 1 superlinearly, but it does not have a definite convergence order.

In fact we have seen that when $p = 1$,

$$\lim_{k\to\infty} \frac{|r_{k+1} - 1|}{|r_k - 1|^p} = \lim_{k\to\infty} \frac{r_{k+1} - 1}{r_k - 1} = 0.$$

As the limit is 0, $p = 1$ is not the convergence order. It seems that the convergence order is higher than 1.

But this is not true. We can prove that for any $p > 1$,

$$\lim_{k\to\infty} \frac{|r_{k+1} - 1|}{|r_k - 1|^p} = \infty. \tag{8}$$

So, the sequence also cannot have a convergence order $p > 1$.

We now prove (8). The sequence satisfies:

$$r_2 - 1 = \frac{1}{2}(r_1 - 1) = \frac{1}{2} \cdot 3,$$

$$r_3 - 1 = \frac{1}{2^2}(r_2 - 1) = \frac{1}{2} \cdot \frac{1}{2^2} \cdot 3,$$

$$\cdots\cdots$$

In general,

$$r_{k+1} - 1 = \frac{1}{2} \cdot \frac{1}{2^2} \cdots \frac{1}{2^k} \cdot 3.$$

Let

$$\alpha_k = \frac{|r_{k+1} - 1|}{|r_k - 1|^p} = \frac{r_{k+1} - 1}{(r_k - 1)^p} \quad (p > 1).$$

Then we may obtain

$$\alpha_k = \frac{(2 \cdot 2^2 \cdots 2^{k-1})^{p-1}}{2^k} \cdot 3^{1-p}.$$

As

$$2 \cdot 2^2 \cdots 2^{k-1} = 2^{1+2+\cdots+(k-1)} = 2^{\frac{k(k-1)}{2}},$$

$$\alpha_k = 2^{\frac{k(k-1)}{2}(p-1)-k} \cdot 3^{1-p}.$$

Since $p > 1$,

$$\frac{k(k-1)}{2}(p-1) - k = k[\frac{k-1}{2}(p-1) - 1] \to \infty.$$

Therefore,

$$\alpha_k \to \infty.$$

So far we have introduced the concept of convergence order for a sequence of numbers $\{r_k\}$. For a sequence of points (vectors) x^k in R^n converging to a point (vector) x^*, we may define convergence order of $\{x^k\}$ by the convergence order of the sequence $\{\|x^k - x^*\|\}$. For example, we say that x^k converges to x^* superlinearly if $\|x^k - x^*\|$ converges to 0 superlinearly, i.e.,

$$\lim_{k\to\infty} \frac{\|x^{k+1} - x^*\|}{\|x^k - x^*\|} = 0.$$

Later we shall see that **different algorithms may have different convergence orders**. Here for the single-variable Newton's method and secant method, we give their convergence orders.

Theorem 1. Let function f have a continuous second order derivative, and let x^* satisfy $f'(x^*) = 0$, $f''(x^*) \neq 0$. Then, provided x^0 is sufficiently close to x^*, the sequence $\{x^k\}_{k=0}^{\infty}$ generated by Newton's method converges to x^* with an order of convergence at least two.

Theorem 2. Let function f have a continuous second order derivative, and let x^* satisfy $f'(x^*) = 0$, $f''(x^*) \neq 0$. Then, provided x^0 and x^1 are sufficiently close to x^*, the sequence $\{x^k\}_{k=0}^{\infty}$ generated by the secant method converges to x^* with order $\tau_1 \approx 1.618$.